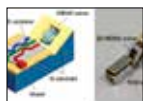




New Snake Species

Scientists discover new species of dwarf burrowing snake - *Levitonius mirus*, in the islands of Philippines. <http://digit.in/jan21-07>



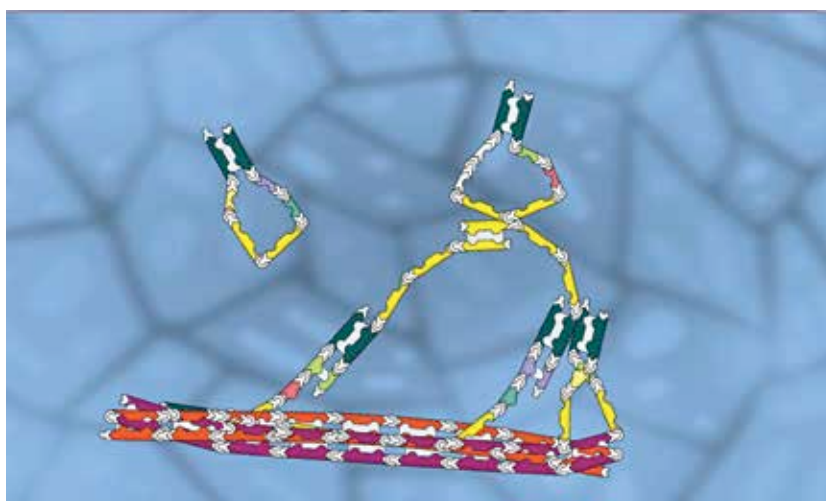
Streaming To The Retina

Japanese researchers develop miniature laser projector that beams HD video straight to the retina. <http://digit.in/jan21-08>

crucial component of future biological circuits that can be used for a wide range of therapeutic and diagnostic purposes in humans. These include systems that can initiate gene expressions at particular points in the circadian cycle, or a timer to check how much time has elapsed since the detection of a biomarker.

In fact, programmable bacteria are already used for cancer therapies. These are all common and benign bacteria that are modified to release therapeutic drugs at the site of a tumour, to make sure that the drugs are not delivered elsewhere. The tumour is detected using simple molecular structures on the surface of the bacteria, or through logic gates programmed into it. The bacteria are considered to be “platforms” that can be activated through chemical, biological, or electromagnetic inputs. Note that most of the medical apparatus is contained within the bacteria.

Another approach uses modified T cells introduced into the body, to improve the immune system. Known as chimeric antigen receptor (CAR) treatment, the T cells can detect and attack cancer cells. The therapy is designed to accentuate the immune



An example of a DNA machine made in Nanocrafter

system, with the modified T cells having kill switches that neutralise the cells in case the patient has an adverse reaction. These examples are just the earliest and most rudimentary examples of programmable medicine, and there is plenty of scope for advancements in this domain.

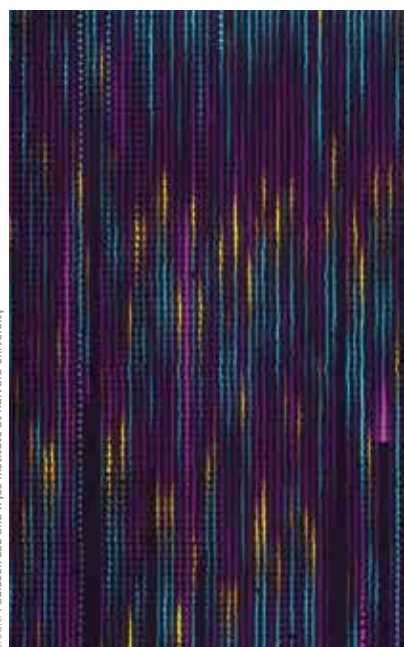
There are some therapeutic avenues that scientists are still exploring, such as the possibility of genetically modifying viruses for therapeutic uses. In some experimental procedures, genetically modified viruses are themselves used to reprogram genes in patients with inherited diseases. Another area of research is using plant material to induce antibody production, which would be a cheap edible vaccine that can be grown from the ground, to combat a future pandemic. There is also significant interest in investigating the therapeutic value of stem cells. In theory, it is possible to program somatic cells that produce the relevant stem cells on demand, which can then repair damaged tissue in the body or even regrow missing organs.

Researchers from Georgia Tech are focusing on biocircuitry research that focuses on making programmable drugs. Though these drugs are still in the theoretical stage, when they are eventually realised, they are expected to look exactly like conventional medicines. They will be delivered

in similar containers, with the same modes of administration. These medicines would become extensions of our bodies, persisting for long periods of time, delivering appropriate doses of drugs when needed, where they are needed. The drugs can be used to treat cancer, fight infections, and detect diseases. The team began to develop simple logic gates, and then realised that they could make increasingly complex computers, including an analog to digital converter for recording bacterial activity.

The researchers also realised something exciting during the course of their works. Instead of working with “digital” biological circuits that used ones and zeros, they could also make analog circuits that used a wide range of values. These signals can have multiple values between zero and one. The researchers at the lab are working towards building so called “enzyme computers”.

The team developed a test biological circuit, one that could float freely and was not contained within a cell. This circuit could measure the presence of bacteria in blood, and then deliver the appropriate amount of antibacterial drugs. The research shows the promise of using biological circuits, but thinking them as another avenue of digitising, and building up the capacity to work with analog signals. 



Tracking the repressilator cycles across individual bacteria samples

Credit: Paulsson Lab and Wyss Institute at Harvard University

Credit: Paulsson Lab and Wyss Institute at Harvard University